MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS)

THIRDSEMESTEREXAMINATION

(2024ADMISIONONWARDS)

MG3DSEECT202SensorsandActuatorsinIndustryAutomation

Duration:2hours **MaximumMarks: 70**

Studentsshouldattemptatleastonequestionfromeachcourseoutcometoenhancetheiroverall *outcomeattainability.*

**Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S),Interest (I)andAppreciation(Ap)*

PartA

MultipleChoiceQuestions

Answer **All** Questions

Eachquestioncarries1 mark

1.Whichis thebestdescriptionforasensor?

A) Adevicethatstoreselectricalenergy

B) Adevicethat converts electricalsignalsintomechanicalmovement

C) Adevicethatdetectschangesinphysicalconditionsandconvertsthemto signals

D)Adevicethat increasesvoltage 1 U

2.Outofthefollowingstatements, isNOTafunctionofan actuator?

A) Controllingmechanicalsystems

B) Generatingsignalsfromthe environment

C) Movingorcontrolling a mechanism

D) Executinga commandfromacontroller 1 K

3.Which ofthe followingisatemperaturesensor?

A)LDR B)Thermocouple

C) Microphone D)Ultrasonic sensor 1 U

4. Aproximity sensorisusedto detect:

A) Light B) Sound

C)Nearby objects D)Temperature 1 K

5. InwhichapplicationisanIRsensornottypicallyused?

A) Gasleakage detection

B) TVremote control

C) Obstacledetection

D) Linedetection 1 K

6.What isatransducer?

A) Adevicethatstoresenergy

B) Adevicethatamplifiesasignal

C) Adevicethat convertsoneformofenergyintoanother

D) Adevicethatdisplays output 2 U

7.Thetransducerthatrequiresanexternal powersourceforoperationis,

A)Activetransducer B) Passive transducer

C)Digitaltransducer D)Output transducer 2 K

8. Whichinstrumentistypicallyconnectedacrossthebridgeforbalance detection?

A) Ammeter B)Voltmeter

C) Galvanometer D) Multimeter 2 U

9. Interfacingoftransducers means,

A) Connectingpower supplyto transducer

B) Connectingtransducersdirectlytotheload

C) Connectingtransducerstosignalconditioningandprocessingcircuits

D) Replacingtransducerswithactuators 2 K

10.WhichADCisbestforlow-speedandhigh-accuracymeasurements?

A)FlashADC B)Dual-slopeADC

C) SARADC D)R-2RladderDAC 2 K

11.Themainfunctionofanactuatorina controlsystemis,

A)Receivefeedbackfromenvironment

B) Measuresignalvalues

C) Executecontrolaction

D) Convertanalog signalto digital 3 U

12.Whatisthe mainadvantageof electricactuatorsover pneumatic andhydraulicones?

A)High forceoutput B)Lowcost

C)Cleanandpreciseoperation D)Requiresalargecompressor 3 A

13.Whichtypeofactuatoris bestsuitedforexplosiveorflammableenvironments?

A)Electric actuator B)Pneumatic actuator

C)Hydraulic actuator D)Steppermotor 3 A

14.Whatisthefunctionofa drivercircuit ininterfacingactuatorswithmicrocontrollers?

A)Tostoredata B)Toreducecurrent

C)Toamplifycurrentorvoltage D)Toconnectsensors 3 An

15.Inindustrialprocesslines,linearactuatorsareusedto:

A)Maintainairpressure B)Logtemperaturedata

C)Openandclosevalves precisely D)Amplifyanalog signals 3 K

16.Inindustrialautomationarchitecture,whichlayer handlesthedirectcontrolofmachines?

A)Field level B)Control level

C)Supervisorylevel D)Enterpriselevel 4 U

17.Themostcommonlyused controllerinindustrialautomationis,

A)PersonalComputer B)PLC(ProgrammableLogicController)

C) Microphone D)Server 4 U

18. Whichsensorissuitable fornon-contactspeedmeasurement?

A) Potentiometer B)Photoelectric sensor

C) LVDT D)Loadcell 4 K

19. Capacitivelevelsensorsarebestsuitedfor:

A) Measuring speed B)Detectinggas leaks

C) Measuringlevelofnon-conductive liquids

D) Measuring temperature 4 U

20. Whichofthefollowingsensorsiswidely usedinindustrialautomationfor objectdetection

A)Proximity sensor B)Thermistor

C)Gas sensor D)Soil moisturesensor 4 U

20*1=20

PartB

ShortAnswerQuestions

Answerany**6** Questions

EachQuestioncarries5marks

21. Whatarethe differenttypesofsensors andexplainthembasedonthe quantitythey measurewith suitableexamplesforeach type. 1 U

22.Mentionabouttheroleof instrumentation amplifiersinsignal conditioning. 1 K

23.What isatransducer?Explainbriefly aboutWheatstone bridgecircuits. 2 U

24.Writeshortnotesontheinterfacingof transducers. 2 U

25.Withexample,explainbrieflyaboutdifferenttypesofactuators. 3 U

26.Writeshortnoteson(a)relayand(b) solenoid. 3 A

27.Whatarethedifferenttools thatareneededinindustrialautomation? 4 K

28.Explainbriefly abouthumidity sensor. 4 U

6 * 5 = 30

PartC

Essay

QuestionsAnswerany**2**Q

uestions

EachQuestioncarries10marks

29. Whysignalconditioners areneededforsensors andtransducers.Mentionabouttheroleof instrumentationamplifiersinsignalconditioners. 1 U

30. Explainwhyconvertersareneeded.Explainindetailaboutsuccessiveapproximationmethod ADC. 2 U

31. Explainin detail about thefollowing actuators,

A)Servomotor

B)Solenoid 3 An

32.Describeindetailabout differenttoolsusedinindustrialautomation. 4 U

2 * 10 = 20